

850nm vs 940nm, Specs & Applications (Tech-LED)

 tech-led.com/near-infrared-nir-led

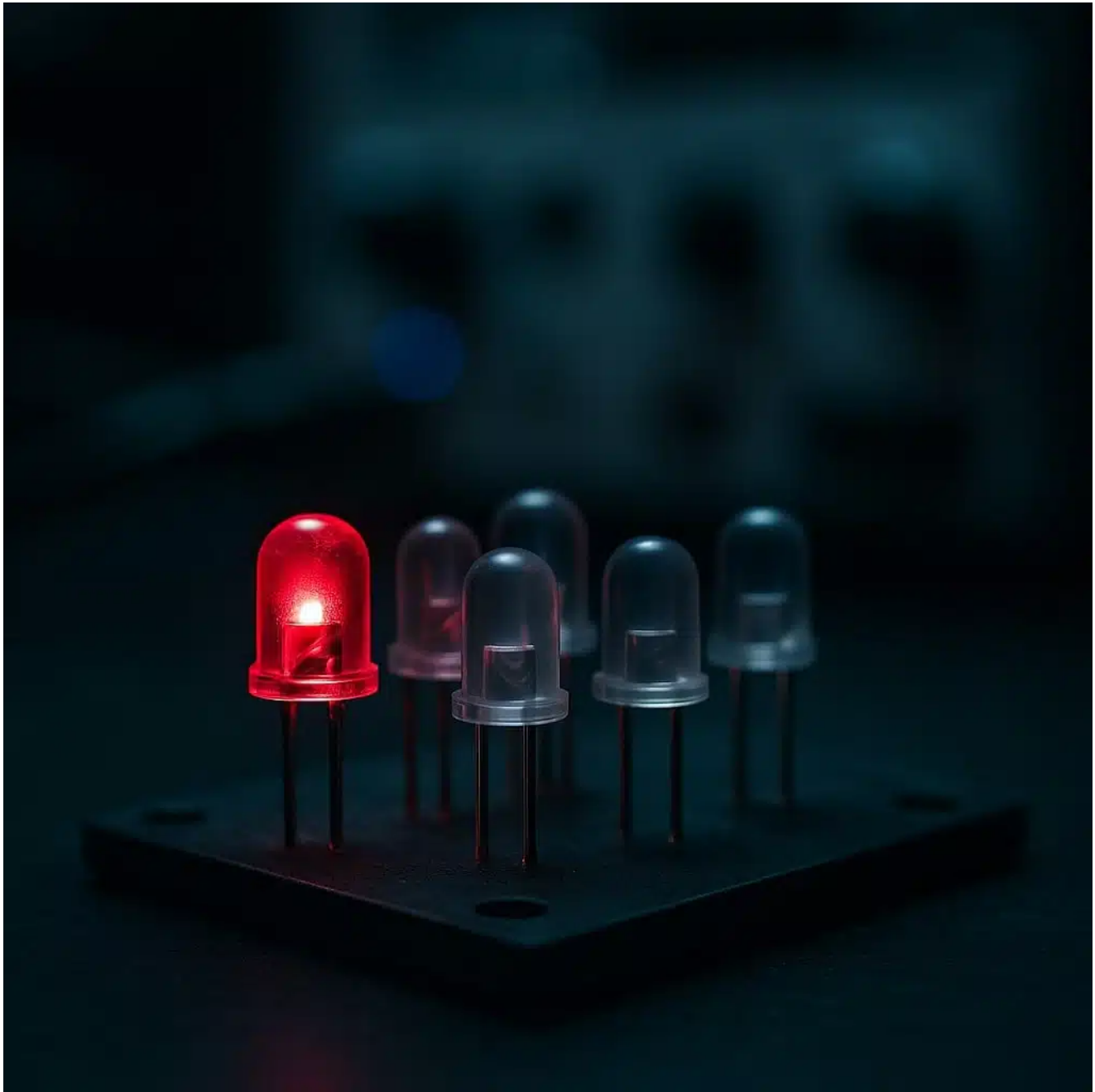
Tech Led

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What Is a Near-Infrared LED?

A [near-infrared \(NIR\) LED](#) is a light-emitting diode that outputs invisible infrared light typically in the 700 nm to 1000 nm wavelength range, just beyond the deep red portion of the visible spectrum. Like any LED, it's a semiconductor device: when forward-biased, electrons and holes recombine to emit photons of light. The semiconductor's bandgap energy determines the photon wavelength – in this case, a longer wavelength in the IR range rather than visible light. Common NIR LED die materials include GaAs (gallium arsenide) and AlGaAs (aluminum gallium arsenide) for wavelengths around 800–890 nm, and InGaAs (indium gallium arsenide) for extending toward 940 nm and beyond^[1]. NIR LEDs often appear dark or faintly red to the eye, since the bulk of their emission is outside human vision.

Because their output is invisible IR radiation, NIR LEDs are used in situations where illumination or signaling needs to be undetectable to humans but detectable to electronics. These range from TV remote controls at the longer end (~940 nm) to covert security camera illuminators. The packages can look like standard visible LEDs (often with a tinted or clear lens) or as SMD/high-power emitters with a black-tinted epoxy to filter any tiny visible component. Marubeni Tech-LED, a division of Marubeni America Corporation, specializes in such IR emitters and offers a broad lineup of NIR LED products for different applications (details in a later section).



How NIR LEDs Work

Electrically, a near-infrared LED works just like a visible LED – by electroluminescence. When current flows through the diode’s p-n junction, electrons drop into lower-energy holes and release energy as photons. The photon wavelength (λ) is roughly given by $\lambda \approx (1.24 \mu\text{m}\cdot\text{eV})/E_g$, where E_g is the semiconductor bandgap in electronvolts. For example, AlGaAs with a suitable composition might have $E_g \approx 1.45 \text{ eV}$, yielding $\sim 855 \text{ nm}$ emission (near IR). This means the LED’s material science is tuned such that its output lands in the IR-A band (also called near IR or IR-A, $\sim 700\text{--}1400 \text{ nm}$). In practice, most “near IR” LEDs on the market cover roughly 730 nm up to about 950 nm before LED efficiency drops off and other technologies (like laser diodes or SWIR emitters) take over.

Wavelength Bands (700–1000 nm) & Specs

Near-infrared LEDs generally span the IR-A sub-band of the infrared spectrum (often defined as 700–1400 nm), but LED emitters are most common up to ~1000 nm due to material constraints. Within ~700–1000 nm, typical LED peak wavelengths include 730 nm, 850 nm, 880 nm, 940 nm, and a few extending to ~970–1050 nm. It's useful to note that these wavelength values are approximate – each LED has a spectral bandwidth (full width at half maximum, FWHM) on the order of 25–50 nm. For example, an “850 nm LED” might have a peak output around 850 nm, but actually emits a broad band roughly 835–865 nm (FWHM ~40 nm). This broad output is a key difference from laser diodes, which emit at very narrow wavelengths. The broader bandwidth can be advantageous for certain sensors (e.g. avoiding interference patterns), but it means the light is less monochromatic.

Typical Electrical & Optical Specs: [Near-IR LEDs](#) share many electrical characteristics with their visible-light cousins. Forward voltage (V_F) is usually in the ~1.2–1.8 V range at nominal current, since the bandgap is lower than that of visible LEDs. For instance, a standard 850 nm LED might have $V_F \approx 1.5$ V at 50 mA. Because many IR LEDs use high-efficiency GaAlAs chips, they can handle relatively high currents; DC drive currents of 50–100 mA are common for discrete IR LEDs, with pulsed currents up to ~1 A for short bursts (to extend range, e.g. in remote control transmitters). In terms of radiant output, a single IR LED typically emits on the order of tens of milliwatts of radiant power. For example, Tech-LED's SMT850-25 (a top-emitting 850 nm LED) produces about 16–22 mW of radiant power at 50 mA^[2], or up to ~44 mW when pulsed at 100 mA^[3]. Likewise, a comparable 940 nm LED might emit ~10–20 mW at 50 mA (a somewhat lower output due to material efficiency and eye-invisibility tradeoffs). Table 1 summarizes some typical performance ranges for near-IR LEDs:

Parameter	850 nm LED Example	940 nm LED Example
Peak Wavelength	~850 nm (NIR band)	~940 nm (NIR band)
Spectral Bandwidth	~30–40 nm FWHM	~40–50 nm FWHM (typical)
Forward Voltage (V_F)	~1.5 V @ 50 mA	~1.2–1.4 V @ 20 mA (typ.)
Radiant Power	~20–40 mW @ 50–100 mA	~15–30 mW @ 50–100 mA (typ.)
Radiant Intensity	~80 mW/sr (with narrow lens)	~50 mW/sr (with typical lens)
Viewing Angle ($2\theta_{1/2}$)	~±20° (narrow lens)	~±30° (typical lens)
Package Types	3–5 mm through-hole, SMD, COB	3–5 mm through-hole, SMD, COB

Table 1. Typical Specifications for Near-Infrared LEDs (example values)

Note: 850 nm devices (often AlGaAs-based) tend to have slightly higher output than 940 nm devices, partly due to material efficiency and human-eye invisibility tradeoffs. Engineers should note that these values vary by product and manufacturer. Modern high-power IR LEDs (often in SMT packages or chip-on-board arrays) can achieve much higher outputs. For instance, Luminus quotes up to 10–15 W continuous optical output for their largest IR LED arrays, covering wavelengths from 730 nm up to 940 nm – a testament to how far LED technology has advanced into traditional laser domains^[4]. Another important point is that near-IR LEDs produce minimal heat in their output beam (since the radiation is mostly beyond visible and each photon carries lower energy than visible or UV). Most heat generated is due to electrical inefficiency and must be dissipated through the LED’s thermal pad or lead frame (proper thermal design is covered in a later section). For a deeper exploration of how [NIR technology drives sustainability](#) in smart buildings, industrial systems, and off-grid applications, see our article on [NIR LEDs for Energy Efficiency & Sustainability](#).

Industrial / Medical Applications

Near-infrared LEDs are used in a remarkably broad range of applications, especially where invisible illumination or IR sensing is needed. [Key industries and use-cases include:](#)

- **Machine Vision & Industrial Automation:** NIR LEDs provide illumination for cameras in machine vision systems (e.g. inspecting parts on a production line) without adding visible glare. Many machine vision cameras have enhanced sensitivity in the 800–900 nm range. IR LED lighting can reveal surface defects or contrast that visible light cannot, and it doesn’t distract human operators. NIR can also penetrate certain materials (like semiconductor silicon or various coatings) to inspect underlying features^[5]. In industrial automation, IR emitters are also used in break-beam sensors, optical encoders, and LiDAR-like distance sensors (some time-of-flight sensors use 850–940 nm LED pulses for ranging).

- **Security & Surveillance:** One of the most widespread uses of NIR LEDs is as IR illuminators for CCTV and night-vision cameras. Almost every security camera that offers night vision has a ring of 850 nm or 940 nm LED emitters that flood the area with IR light. At night, the camera's IR-cut filter is removed, allowing it to see the scene lit by infrared as a monochrome image. 850 nm LEDs are commonly used in security cameras because they provide better range and camera sensitivity^[6] (albeit with a faint red glow visible on the LED). Meanwhile, 940 nm LEDs are used for covert surveillance when absolute stealth is required – they emit no visible glow, though at the cost of roughly 50% shorter illumination range and reduced camera sensor response^[7]. A real-world case study is the TOA “TRIFORA” series of outdoor security cameras, which adopted high-power 940 nm LED arrays from Stanley Electric to achieve invisible 0 lux night-vision illumination for critical infrastructure monitoring^[8]. These IR LEDs allow surveillance in total darkness without alerting subjects or drawing unwanted attention.
- **Medical Devices & Life Sciences:** Near-IR light penetrates human tissue to a degree, enabling various biomedical applications. Pulse oximeters use a pair of LEDs (typically one red ~660 nm and one IR ~940 nm) to measure blood oxygen saturation by comparing light absorption. Vein finders and vascular imaging devices often use 850 nm or 880 nm LED illumination – hemoglobin in blood absorbs IR, so shining NIR light and using a camera can reveal sub-surface veins (for easier IV access) as dark lines^[9]. In clinical diagnostics, IR LEDs are used in spectroscopy (e.g. analyzing tissue or fluids) and in medical imaging systems for illumination. Another growing area is IR LED therapy devices (e.g. for muscle pain relief and dermatology treatments), typically using high-power LEDs around 810 nm or 850 nm – we'll address these in a health and wellness context. Even consumer wearables and fitness trackers use NIR LEDs for heart-rate sensing (via photoplethysmography around ~ IR wavelengths).
- **Consumer Electronics:** The humble TV/DVD remote control is a classic example – a 940 nm IR LED blinking a data pattern to a receiver. Beyond that, IR proximity sensors on phones (for gesture recognition or face presence detection) use IR LED emitters paired with photodiodes. Eye-tracking in VR headsets or AR glasses often uses small 850 nm micro-LEDs around the lenses, so that inward-facing cameras can track pupil movements even in the dark. Facial recognition systems (like those in phones or door access devices) might flood the user's face with 850 nm light to get a detailed IR image or to project an IR dot matrix (though structured-light dot projectors are usually IR lasers). Automotive driver-monitoring systems similarly use NIR LEDs for eye tracking at night^[10].

- **Scientific & Specialty:** In agriculture, IR LEDs combined with cameras help assess plant health (NDVI imaging, since healthy vegetation reflects NIR strongly). In astronomy or wildlife observation, IR LED illumination allows viewing nocturnal animals or celestial objects with IR-sensitive equipment. NIR LEDs are also used in optical communications for short-range links; for example, some fiber optic systems around 850 nm use LED sources (though lasers are more common for longer distances). Even art restoration benefits from NIR: IR LED lighting can be used in IR reflectography to see underdrawings in paintings without damaging them.

Mini Case-Study – Machine Vision in Pharmaceuticals: A pharmaceutical packaging line deployed NIR LED line lights to inspect sealed pill blister packs. The near-IR light could penetrate the plastic film and paper backing slightly, revealing if a pill was missing or broken inside its cavity – something not detectable under visible light. An IR-sensitive camera captured the transmitted IR image, allowing the machine vision system to reject defective packs. This real-world example shows how NIR LEDs can “see through” materials and enable quality control that’s invisible to the human eye. The result was a significant drop in packaging defects, without needing X-rays or other expensive systems.

In summary, near-infrared LEDs have become a foundational technology across industries – from keeping us safe (security, automotive) to enabling innovative medical diagnostics – all by delivering illumination and sensing at wavelengths we can’t see, but our devices can.

Eye-Safety & IEC 60825

One might assume that because near-IR light is invisible, it’s harmless to eyes. In fact, the opposite is true: infrared radiation in the 700–1400 nm range can pose a significant hazard to the retina precisely because it triggers no blink reflex or pain while being focused on the retina (this range is sometimes called the “retinal hazard region”). High-power IR LEDs or lasers can cause retinal burns or cataracts without a person realizing they’ve been exposed. As a result, safety standards typically treat high-output IR LEDs with the same caution as lasers.

Laser Safety Standards for LEDs: The international standard IEC 60825-1 (“Safety of Laser Products”) historically covered LEDs as well as lasers, classifying them by hazard class. According to Lasernet, the IEC 60825-1 standards apply equally to lasers and LEDs, and generally most LEDs fall into lower classes (Class 1, 1M, 2, etc.)^[11]. Practically speaking, most small IR LEDs are Class 1 devices – meaning eye-safe under all conditions – because their power and viewing aperture are limited. Vishay’s eye-safety testing found that none of their individual IR emitters violated the Class 1 limit, and under the newer IEC 62471 “lamp safety” standard, almost all were in the Exempt Group (no hazard)^[12]. However, arrays of IR LEDs or very high-power emitters can reach Class 1M or higher, where using optical instruments (like binoculars) to view them could be hazardous. Manufacturers of IR LED

products must either certify under the laser product standard (IEC 60825-1) or under IEC 62471 (the LED lamp safety standard), and provide maximum permissible exposure distances or safety labels if required.

Practical Eye Safety Tips: Engineers integrating NIR LEDs should follow some basic precautions:

- Never stare into high-power IR LEDs at close range, even if they appear dim or off. Your eyes won't detect the IR, but the energy could still be focusing on your retina.
- Use IR viewing cards or IR cameras to check IR illumination instead of your eyes. These tools can visualize the IR beam so you know if it's active.
- Enclose or shield powerful IR LED arrays in products so that end-users can't easily bypass diffusers or covers and stare directly at the source.
- Follow Class 1 LED emission limits for consumer products. Aim to keep the IR output below the accessible emission limits for Class 1 as defined by IEC 60825-1/IEC 62471, especially for wavelengths that are invisible (the limits for IR are quite strict due to the lack of aversion response).
- If using extremely high-power IR illumination (e.g. IR lamps for military or large-area lighting), treat it as hazardous as a laser. Operators might need to wear IR safety goggles that filter out near-IR (these can look clear but have IR-blocking coatings).

It's worth noting that outside the 400–1400 nm range, the concern shifts to corneal and lens damage (since longer-wave IR is absorbed in the front of the eye). But a near-IR LED definitely falls in the retinal hazard band. Fortunately, as mentioned, most off-the-shelf IR emitters for electronics are low-power. For example, a typical 5 mm IR diode in a remote control is orders of magnitude below eye-hazard levels – it's when you start ganging dozens of high-power LEDs or driving them at high current that safety classifications become relevant. Always consult the LED's datasheet and IEC 60825-1 classification if provided. Marubeni Tech-LED provides guidance on safe usage for its high-output IR LED clusters; for any custom high-power IR illumination, it's wise to perform an irradiance measurement at relevant distances to ensure compliance.

In summary, treat near-infrared light with respect. The same invisibility that is so useful in applications can also make it dangerous. Adhering to standards like IEC 60825 and IEC 62471 and implementing simple safety measures will ensure your NIR LED-based design is both effective and safe.

Tech-LED Product Line (SMT850, 940 nm Arrays) + Datasheet Links

Marubeni Tech-LED offers a comprehensive lineup of near-infrared LED products, spanning from single-die components to multi-chip arrays. As a division of Marubeni America Corporation focused on optoelectronics, Tech-LED's portfolio addresses the needs of

machine vision engineers, medical device designers, and security system integrators alike. Here we highlight a few key product series relevant to 850 nm and 940 nm NIR LEDs:

SMT850 Series (850 nm SMT LEDs)

The SMT850 family consists of high-performance 850 nm infrared LEDs in surface-mount packages. For example, SMT850-25 is a top-emitting IR LED with a built-in lens, delivering ~44 mW of radiant power and ~80 mW/sr intensity at 100 mA pulse drive^{[2][3]}. These devices use an AlGaAs chip optimized for 850 nm output. Variants in the series offer different lens options (for various beam angles) and power bins. The SMT850D model is a particularly high-output version in a compact package – featuring a single 350 × 350 μm AlGaAs die but with enhanced thermal management, allowing up to 100 mA continuous drive^[14]. These LEDs are ideal for IR illumination in compact devices like gesture sensors, CCTV camera IR boards, or medical sensor probes. (Internal link suggestion: include a link to the SMT850 series datasheet PDF or product page.)

SMT940 Series (940 nm SMT LEDs)

Complementing the 850 nm parts, Tech-LED's SMT940 series are 940 nm IR LEDs in similar SMT packages. They use a GaAlAs/GaAs chip tuned for 940 nm. While they have slightly lower radiant intensity than their 850 nm siblings, they are completely invisible (no red glow) – perfect for covert surveillance or any application where even faint visible light must be avoided. For instance, SMT940-25 would denote a lens-type version akin to the 850-25 but at 940 nm, with typical outputs on the order of ~30 mW at 50 mA (datasheet values). SMT940D variants exist as well, indicating high-power versions (higher current capability). (Link to SMT940 series datasheet or product page.)

Multi-Chip IR LED Arrays (850 nm & 940 nm)

For applications needing higher power IR illumination – such as long-range night vision, machine vision lighting, or medical therapy devices – Tech-LED offers multi-die IR LED arrays. These include products in metal can packages like the TO-66 IR LED arrays. For example, part L850-66-60 is an array of 60 chips emitting at 850 nm, mounted in a TO-66 metal can header with a flat glass window. It can handle up to ~600 mA forward current with a combined forward voltage around 17–20 V (since many chips are series-connected). Such an array can output on the order of hundreds of milliwatts of IR power. (Tech-LED's catalog lists ~130 mW radiant flux for a 60-chip UV version at 365 nm as a reference^[15], and the IR versions achieve even higher output given the better LED efficiency in IR.) Similarly, L940-66-60 is a 60-chip 940 nm array in a TO-66 package – essentially an IR flood lamp in LED form – useful for high-end security illuminators, IR searchlights, or large-area machine vision in warehouses. Tech-LED also provides IR cluster modules (denoted by part codes like

SMBB or Lxxxx) which incorporate multiple LED dice on a substrate or board, for instance a 940 nm 16-chip SMT cluster that can be easily mounted on a heatsink. (Link to high-power IR LED arrays datasheet or catalog page.)

Custom Wavelength and Multi-Wavelength LEDs

Beyond the standard 850 nm/940 nm options, Tech-LED's portfolio (inherited from the Epitex line) covers other near-IR wavelengths such as 810 nm, 880 nm, 905 nm, etc., and even multi-wavelength emitters. For example, part SMT850/940 combines an 850 nm and a 940 nm LED die in one package, allowing dual-band operation. Tech-LED also offers IR VCSEL solutions and matching photodiodes to pair with their LEDs, providing complete IR emitter-detector solutions.

Each product line comes with a detailed datasheet specifying radiant intensity curves, spatial radiation patterns, and handling precautions. Engineers can download our 850 nm & 940 nm datasheet pack (which includes the latest Tech-LED IR emitter datasheets) for exact specs and integration notes – this is highly recommended during design to ensure proper drive circuitry and optical layout. ([Download 850 nm & 940 nm Datasheet Pack.](#)) Additionally, our website features a wavelength search page where you can filter LED products by peak wavelength (for example, see all ~850 nm options or all ~940 nm options in one list). This is a handy way to explore available emitters for your required spectral band.

Lastly, Tech-LED stands ready to assist with custom solutions – for instance, if an application demands an 870 nm LED array or a specific beam-shaping optic, our team can advise on modifications or custom part numbers. We pride ourselves on being a one-stop source for NIR optoelectronics, backed by Marubeni's quality and supply chain strength.

Design Checklist (Drive Current, Thermal, Optics)

Designing with near-infrared LEDs requires careful consideration of several engineering factors. Below is a checklist for incorporating NIR LEDs (such as 850 nm or 940 nm devices) into your project, ensuring performance and longevity:

Drive Current & Pulsing

Determine the appropriate forward drive current for your NIR LED and how you will drive it (constant current source, resistor, PWM, etc.). Most IR LEDs have a max continuous current (e.g. 50 mA or 100 mA) and a higher allowable peak pulsed current (e.g. 0.5–1 A for very short pulses at low duty cycle). If your application (say, an IR proximity sensor or camera illuminator) can use pulsed IR, you can achieve higher peak output by pulsing within the safe limits – for example, pulsing an 850 nm LED at 1000 mA with 1% duty cycle to extend range. Always use a constant current driver or a current-limiting resistor; driving IR LEDs directly from a fixed voltage source can cause thermal runaway since V_F drops with temperature.

Also consider rise/fall times – IR LEDs switch on and off very fast (nanosecond-scale), so you may want to add a series inductance or an RC snubber if needed to limit EMI from fast edges (especially when pulsing at ~38 kHz for remote controls).

Thermal Management

Even though IR LEDs are efficient, a significant fraction of input power still turns into heat. For example, at 100 mA and ~1.5 V forward, about 0.15 W is dissipated as heat per LED. High-power IR LED modules (especially multi-chip arrays consuming several watts) require proper heatsinking. Check the thermal resistance ($R_{\theta JA}$ or $R_{\theta JC}$) in the datasheet – for instance, it can be around 80 K/W in a small LED, versus as low as ~2 K/W in a TO-66 array package^[15]. Use metal-core PCBs or solder the LED to thermal pads connected to large copper areas. Ensure there is airflow or a metal chassis acting as a heat spreader if needed. Remember that as junction temperature rises, the LED's output power drops and its wavelength might shift slightly. Design for the worst-case ambient temperature and drive current; you may even incorporate thermal derating in your driver (e.g. reduce current if the LED board temperature exceeds a certain threshold).

Optics and Beam Shaping

Decide on the beam requirements for your NIR illumination. Do you need a narrow beam to reach a far target, or a wide flood to cover an area? Many IR LEDs come with built-in lens options (for example, Tech-LED parts may have suffixes like -23, -25, -27 indicating different lens types with $\sim\pm 10^\circ$, $\pm 20^\circ$, $\pm 30^\circ$ half-angles). If using a bare LED chip or a flat-window package, you might need external optics: TIR lenses, LED reflectors, or simple diffusers. For example, machine vision systems often use collimating optics to get a tight IR spot, whereas a security camera might use a diffuser film to spread IR evenly over the field of view. When mixing 850 nm and 940 nm LEDs in one system, note that their emission profiles might differ slightly; you may need to adjust optics or alignment for each. Also account for the materials of any optical elements – many plastics and glasses transmit NIR well, but some coatings (e.g. certain anti-reflection coatings) might be optimized for visible light and not IR. Use IR-transparent materials (acrylic, polycarbonate, quartz, etc., with known transmission at your wavelength) for any lens or cover in front of the LED.

Power Supply and Switching

IR LED drivers should prevent large current overshoot. If you're driving an array of 60 IR LEDs in series (perhaps ~20 V forward drop), make sure your driver can handle that voltage plus some overhead and remains stable. Pay attention to the voltage-temperature coefficient of the LED's forward drop (typically around -2 mV/°C per diode for silicon, but for IR LED compounds it might be on the order of -1 to -3 mV/°C). This means as the LED heats up, its V_F drops, which in a simple resistor-limited circuit would increase current – another reason

constant current regulation is preferred. If multiplexing or syncing LED pulses with camera exposure (common in machine vision, to strobe IR in sync with the shutter), ensure timing is precise and that the LED can handle the chosen duty cycle.

Eye Safety & Regulations

As discussed in the safety section above, incorporate design features to ensure user safety. For example, if you have a high-power IR lamp in a product, you might include a small visible indicator LED or other signal to show when the IR illuminator is on (since you can't see the IR beam itself). Additionally, consider regulatory classifications: if your device will be sold on the market, you may need to include an IEC 60825-1 laser hazard label or a statement in the user manual if the IR output exceeds Class 1 eye-safe limits. For most small designs (like a single SMD IR LED for a proximity sensor), this won't be an issue, but it becomes important for larger illuminators and lighting products.

Environmental Considerations

NIR LEDs intended for outdoor or industrial use should have proper encapsulation. Many IR LEDs come in epoxy lenses which can yellow under years of sun UV exposure; consider UV-resistant encapsulants or placing the LED behind an IR-pass filter that also blocks UV. Check the operating temperature range – typically -40 to +85 °C is standard for IR LEDs, which covers most environments. If using IR LEDs in a pulsed high-current mode, verify their solder joint reliability (thermal cycling from repeated pulses can stress solder joints – following the recommended PCB footprint and soldering guidelines helps). In general, design for ruggedness if the LEDs will see wide temperature swings or harsh conditions.

Matching Detectors

Often an IR LED is only half of a system – it's paired with a photodiode, phototransistor, or camera sensor. Ensure your detector is sensitive at the LED's wavelength. Silicon photodiodes and typical CCD/CMOS camera sensors have good sensitivity up to ~950 nm, usually peaking around 800–900 nm. Beyond ~1000 nm you'd need InGaAs detectors. So 850 nm and 940 nm are fine with common silicon-based sensors (though note that silicon's quantum efficiency at 940 nm is a bit lower than at 850 nm). Also, if ambient light is a concern, you might want an IR band-pass filter on the detector that matches your LED (for instance, an 850 nm ± 20 nm filter to cut out other light). In communication applications, consider data rate – IR LEDs can generally be modulated at several MHz (as seen in IrDA or remote controls), but check rise/fall times if pushing into high-frequency modulation.

By checking off these considerations – drive current, thermal management, optical design, safety, and sensor integration – you'll greatly increase the chances of first-pass success with your near-infrared LED design. In essence, treat an IR LED design with the same rigor as a

visible LED or even a laser design: manage the heat, control the current, shape the light, and respect the hazards.

850 nm vs 940 nm Comparison

When selecting a NIR LED, a frequent question is “850 nm or 940 nm?” These two wavelengths are by far the most commonly used in IR emitters. Each has distinct advantages and considerations. We’ll compare them across several factors:

Radiant Intensity & Range

850 nm LEDs typically emit roughly 2–3× the radiant intensity of 940 nm LEDs at the same drive current^[16]. This is partly due to material efficiency and partly due to how silicon detectors respond (they tend to have higher quantum efficiency around 850–900 nm than at 940 nm). In practical terms, an IR illuminator built with 850 nm LEDs might reach ~100 m range, whereas a similar-power 940 nm illuminator might only reach ~50–70 m under the same conditions^[16]. So if maximum range or brightness is the goal, 850 nm usually wins. (The illustrative Figure 1 in our PDF compared relative output of the two.)

Visibility (Stealth Factor)

850 nm LEDs do produce a faint cherry-red glow when viewed directly, especially at high power. In dark conditions, an 850 nm security camera illuminator looks like a dim red glow – potentially noticeable if someone is looking at the camera. 940 nm LEDs, on the other hand, are effectively invisible to the human eye – there is no perceptible glow even at high drive levels. This makes 940 nm ideal for covert applications where even a small glow is unacceptable (e.g. military gear, wildlife monitoring, high-end security). For example, in certain environments like railways or airports, 940 nm IR lighting is used so that no red lights could be mistaken for signal lights. If stealth is critical, 940 nm is the better choice.

Camera Sensitivity

Most monochrome or IR-capable cameras use silicon-based sensors that are quite sensitive in the ~800–900 nm range, then sensitivity tapers off toward ~1000 nm. Many off-the-shelf security cameras, for instance, are most sensitive around 850 nm; at 940 nm their sensitivity might drop significantly (to half or less of the 850 nm response)^[17]. This means for a given LED power, a camera will “see” the 850 nm illumination more brightly than the 940 nm. Some specialty cameras (or those with enhanced night-vision sensors) have improved 940 nm response, but generally speaking, sensor spectral response is a point in favor of 850 nm.

Interference and Ambient Light

Interestingly, 940 nm LEDs face slightly less ambient light interference outdoors because sunlight and many artificial lights emit less energy at 940 nm than at 850 nm. (Sunlight has an IR tail that extends beyond 850 nm, but its intensity at 940 nm is somewhat lower relative to shorter IR wavelengths.) This can translate to a small contrast advantage for 940 nm in daylight rejection scenarios (active IR systems, etc.). However, the difference is minor – both 850 nm and 940 nm illumination will be overwhelmed by direct sunlight, and for indoor use neither typically suffers much ambient interference unless there are other IR sources present.

Applications & Ecosystem

The surrounding ecosystem of detectors and accessories can influence the choice. 850 nm has an edge in that many IR-pass camera filters (the kind used on CCTV lenses for night mode) are designed around 850 nm. Also, IR LEDs at 850 nm tend to have more high-power options available (since they are heavily used in CCTV/security, which is a huge market). 940 nm LEDs are extremely common in remote controls and small emitters, but high-power 940 nm arrays are slightly less common (though still available, as we produce them at Tech-LED). If you need to integrate with IR laser components, note that 905 nm and 940 nm laser diodes exist (for LIDAR and rangefinders), whereas 850 nm lasers are relatively rare – so for hybrid systems, you might stick to 905/940 nm for consistency. In machine vision, 850 nm is often preferred because many machine vision lighting vendors produce 850 nm ring lights, line lights, etc., and camera manufacturers optimize for that band. In biometrics (face/iris scanning), 940 nm has become popular in consumer devices to avoid any visible glow (e.g. smartphone face unlock often uses 940 nm flood illuminators).

In summary, the 850 nm vs 940 nm decision boils down to performance vs stealth. If you can tolerate a small red glow and want maximum IR output per LED, choose 850 nm. If total invisibility is crucial and you can compensate for lower output (by using more LEDs or accepting shorter range), choose 940 nm. Many professional security systems actually use a combination: 850 nm for general illumination and a few 940 nm illuminators they can switch on for covert mode. Some devices even allow the end-user to toggle IR wavelength. In either case, Marubeni Tech-LED offers products in both wavelengths (and others in between), and can provide engineering samples of 850 nm and 940 nm LEDs so you can experiment to see which best meets your application's needs. If you need more guidance on [how to choose the correct NIR LED for your business, refer to our guide here](#).

FAQ (People Also Ask)

Do LEDs emit near-infrared?

A: Standard visible-light LEDs produce very little to no near-infrared radiation. Unlike incandescent bulbs which waste a lot of energy as heat and IR, LEDs convert current very efficiently into specific wavelengths of light (e.g. red, green, blue), with almost all output in the visible band – any IR emission is negligible. This lack of IR output is actually a benefit in many applications (for example, LED lighting in museums doesn't damage artwork with IR heat). Only specially designed IR LED devices (with the right semiconductor bandgap) will emit significant infrared. In summary, unless an LED is specified as IR, you can assume a normal LED emits essentially no near-IR. For situations where IR emission is needed (remote controls, sensors), one uses a purpose-built IR LED^[18].

What is the difference between a normal LED and an infrared LED?

A: The primary difference is the wavelength of light emitted. A “normal” LED – say a red, green, or blue LED – emits visible light that we can see, using semiconductor materials with bandgaps corresponding to visible photon energies. An infrared LED uses a different semiconductor (like GaAs or AlGaAs) with a smaller bandgap, causing it to emit photons in the infrared range (typically 800–940 nm) which are invisible to the human eye. Physically, infrared LEDs often look similar to visible LEDs (they might even use a transparent or tinted epoxy package), but sometimes the package is made black or dark blue to act as an IR filter. Electrically, IR LEDs have a lower forward voltage (around 1.2–1.5 V) compared to many visible LEDs (which can be 2–3 V or more) due to that smaller bandgap. In terms of usage, a normal LED provides illumination or indication for humans (indicator lights, displays, lighting), whereas an [IR LED is usually used for](#) communication with devices or for illumination that cameras/sensors can see but people cannot. Also, note that you cannot tell if an IR LED is on just by looking (since it doesn't visibly glow), whereas a normal LED obviously lights up when active. In summary: normal LEDs emit visible light for humans, while IR LEDs emit infrared light for electronic detection.

How efficient is a near-infrared LED?

A: Yes, modern near-IR LEDs are quite efficient in converting electrical power to light – often on par with or even exceeding some visible LEDs in terms of radiative efficiency. Efficiency can be defined in two ways: (1) wall-plug efficiency (radiant flux out vs. electrical power in) and (2) external quantum efficiency (photons out vs. electrons in). Many IR LEDs have wall-plug efficiencies in the range of 30–50%. For example, a high-quality 850 nm LED might output ~100 mW of IR optical power for ~300 mW electrical input (~33% efficiency). This is because the internal quantum efficiency is high and the lower photon energy (IR photons carry less energy than visible photons) means less energy per photon is lost as heat. In practical terms, IR LEDs are very efficient emitters of infrared radiation, and they don't run as

hot as, say, incandescent IR sources. Moreover, [IR LEDs are highly efficient](#) compared to broadband IR lamps, because LEDs concentrate energy at the desired wavelength rather than wasting it across a wide spectrum. As one industry guide notes, IR LEDs convert a high percentage of electrical energy into IR light output^[19]. Additionally, IR LEDs benefit from not stimulating the human eye, so all their output is “useful” for IR sensing (none is visible waste). The exact efficiency will depend on the specific LED model and operating current (efficiency often peaks at a certain current and then droops at higher currents). Generally speaking, NIR LEDs can be considered very efficient IR light sources – which is why they’ve replaced thermal IR emitters in most applications from remote controls to night-vision lighting.

[What are infrared LEDs used for?](#)

A: [Infrared LEDs \(IR LEDs\) are](#) used in a wide array of applications wherever invisible light is beneficial or required. Some of the most common uses include:

- *Remote Controls and IR Communication:* The small IR LED in your TV or DVD remote is a prime example – it sends coded pulses of ~940 nm light to control your television or other appliances. IR LEDs also enabled wireless data transfer in certain devices (IrDA ports on older electronics, though those are much less common now)^[20].
- *Security Camera Illumination:* IR LEDs act as invisible floodlights for CCTV and security cameras, providing night vision illumination. An array of IR LEDs around a camera allows it to “see in the dark” while remaining unobtrusive to people^[21].
- *Surveillance and Military:* Covert IR illumination (often 940 nm LEDs) is used for military night-vision operations and stealth surveillance, since it won’t give away the position of IR spotlights to the naked eye.
- *Sensing and Proximity Sensors:* Many proximity sensors (like the one that turns off your phone screen when you hold it to your ear, or automatic faucet and toilet sensors) use an IR LED paired with a photodiode. The IR LED emits light that reflects off nearby objects and is detected by the sensor. IR break-beam sensors in alarm systems or garage door safety sensors similarly rely on IR LED emitters.
- *Industrial and Machine Vision:* IR LEDs are used to illuminate products on assembly lines for machine vision inspection (the IR can enhance contrast or penetrate materials in ways visible light cannot). They’re also used in optical encoders, line sensors, and other industrial sensing devices.
- *Medical and Biomedical:* As discussed earlier, devices like pulse oximeters and vein finders use IR LEDs. There are also IR therapy devices (850 nm or 880 nm LEDs) used in dermatology and pain relief, as IR light can stimulate tissue and is thought to promote healing in certain cases (a practice known as photobiomodulation).

- *Automotive and Transportation:* IR LEDs are used in some automotive night vision systems (certain cars have IR LED headlights coupled with IR cameras to detect pedestrians/animals at night). They're also found in adaptive cruise control or LIDAR-like systems (though often pulsed laser diodes are used for long range, high-power IR LEDs have seen use in short-range LIDAR). Traffic sensors and license plate recognition cameras might incorporate IR LED illuminators as well.
- *Consumer Electronics & IoT:* Eye-tracking and face recognition systems (e.g. in phones or AR/VR headsets) often use IR LEDs to flood the user's face or eyes with IR light for sensing. Gesture sensors (like those in game consoles or laptops) also use IR LED projectors (Microsoft's Kinect, for example, projected a structured IR light pattern). Additionally, short-distance fiber optic communication links can use IR LEDs (e.g. 850 nm for certain multimode fiber systems).

In summary, IR LEDs are used anywhere we need light for a machine but not for a human – from the simple TV remote to sophisticated night vision and sensing systems. Their ability to provide illumination and data transmission “under the radar” of human eyes makes them incredibly versatile in modern technology.

How does an infrared LED work?

A: An infrared LED works on the same principle as any LED – by electroluminescence in a semiconductor junction^[22]. In an IR LED, however, the semiconductor material is chosen to emit IR photons instead of visible light. Here's a breakdown of how it operates:

- When forward-biased (i.e. when a sufficient forward voltage is applied), electrons in the LED's n-type region gain energy and move across the junction into the p-type region, where they recombine with holes. When an electron falls into a hole (recombines), it drops from a conduction-band energy level to a valence-band level and releases energy in the form of a photon. The wavelength of the emitted photon is determined by the bandgap energy of the semiconductor.
- In an IR LED, the bandgap is relatively small (for example, GaAs has a bandgap of ~1.43 eV at room temperature, corresponding to an IR photon of $\lambda \approx 870$ nm using $E = hc/\lambda$). By adjusting the semiconductor composition (such as incorporating aluminum or indium), manufacturers can tune the bandgap to get desired IR wavelengths (common IR LED peaks include 850 nm, 940 nm, etc.).
- Internally, the LED chip is usually a double-heterostructure that confines electrons and holes in an active region to recombine efficiently. It may also have a reflective substrate or mirror layers to direct as many photons out as possible. The chip is mounted in a package that often includes a lens to help collimate or spread the IR light as needed.

- From an external perspective, you apply a forward voltage (about 1.2–1.5 V) and current flows through the LED; unlike a visible LED, you won't see any light with your eyes, but if you use a camera or IR detector card, you'll observe the LED glowing faintly (the IR output)^[21]. It's essentially a solid-state IR light source with no moving parts, and it can turn on and off extremely fast – switching in the order of nanoseconds, which is useful for data transmission or encoded signals (for instance, the 38 kHz pulses used in IR remote controls)^[13].
- One way to visualize the operation is to think of it as a tiny electronic “light bulb” engineered to emit a specific infrared wavelength. The lack of visible output is purely because the photon energy is below what our eye's photoreceptors can detect – but devices like camera sensors can pick it up easily. Infrared LEDs typically have very long lifetimes (50,000+ hours) and rugged operation, since there's no delicate filament or excessive heat; all the action happens at the quantum level inside the semiconductor crystal.

In summary, an IR LED works by pumping electrical current through a semiconductor junction, causing it to emit light at wavelengths longer than ~700 nm. The underlying physics is identical to that of a normal LED – only the materials differ, so that the emitted photons are in the infrared. It's a beautiful combination of quantum physics and practical engineering that gives us invisible light at the flick of a switch.

Q: What is the difference between 850nm and 940nm in NIR LED products?

A: The main difference is in their wavelength. 850nm LEDs are often used for applications like near infrared light therapy and photobiomodulation therapy, while 940nm LEDs are typically used in night vision applications as they emit light that is invisible to the naked eye.

Q: How do 850nm infrared lights benefit red light therapy?

A: 850nm infrared lights are effective in red light therapy as they penetrate the skin effectively, promoting blood circulation and enhancing cellular function, which can aid in healing and reducing inflammation.

Q: Can I use an ir led strip light for plant growth?

A: Yes, using an 850 nm ir led strip light can be beneficial for plant growth, as certain plants respond well to the near-infrared light spectrum, enhancing their growth and photosynthesis.

Q: What types of applications can benefit from using 850nm and 940nm LEDs?

A: Both 850nm and 940nm LEDs have a range of applications, including near-infrared light therapy, night vision devices, and even in flexible led strip lights for various lighting needs.

Q: How does the 660nm red light compare to 850nm and 940nm in terms of effectiveness?

A: 660nm red light is more effective for surface-level treatments, while 850nm and 940nm penetrate deeper into tissues, making them more suitable for therapeutic applications, such as in near infrared light therapy.

Q: Is it safe to use 850nm and 940nm LEDs for extended periods?

A: Yes, both 850nm and 940nm LEDs are generally considered safe for extended use, especially in therapeutic settings, as they do not produce harmful heat or radiation.

Q: What are some common led products that utilize 850nm and 940nm wavelengths?

A: Common led products include 850nm ir led strips, 940nm LEDs used in night vision cameras, and various led light strips designed for horticultural lighting.

Q: How do I choose the right wavelength for my LED light therapy needs?

A: Choosing the right wavelength depends on the desired outcomes. For deeper tissue penetration and therapeutic effects, 850nm is recommended, whereas for applications like night vision, 940nm is more suitable.

Q: What role do LEDs play in [infrared lighting for health benefits](#)?

A: Infrared lighting from LEDs, particularly in the 850nm and 940nm range, is used in various health treatments, including improving blood circulation, reducing inflammation, and promoting healing through near infrared light therapy.

Q: Where can I find a detailed cost comparison of near-infrared LEDs vs traditional IR solutions?

A: For an in-depth breakdown of initial investment, energy savings, maintenance and ROI, see our [Cost Analysis: Investing in Near Infrared LEDs vs. Traditional Solutions](#).

Q: What kind of carbon reductions can NIR LEDs demonstrate?

A: Real-world examples show that [swapping out traditional IR lamps for NIR LEDs can cut inspection-lighting energy use by up to 90%](#), saving thousands of kilowatt-hours and preventing several metric tons of CO₂ emissions per year. For instance, one factory retrofit reduced its annual IR lighting consumption by 9,000 kWh—equivalent to avoiding roughly 4,500 kg of CO₂.

(For more detailed answers and specific use-cases, see our cluster articles on IR LED basics and applications.)

<https://tech-led.com/use-cases-of-led-light-infrared-lower-carbon-emi>

Footnotes

1. [Moon LEDs – The Ultimate Guide to IR LED](#) (common IR LED semiconductor materials) ↵
2. [Tech-LED – SMT850-251 850 nm LED Datasheet](#) (16–22 mW output @ 50 mA) ↵
3. [Tech-LED – SMT850-251 850 nm LED Datasheet](#) (≈44 mW output @ 100 mA pulse) ↵
4. [Luminus – Infrared \(IR\) LEDs Product Page](#) (IR LED arrays up to 10–15 W, 730–940 nm range) ↵
5. [Vision Systems – Machine Vision Lighting for Pharmaceutical and Medical Device Inspection](#) (NIR penetration of materials for inspection) ↵
6. [Axton Tech – 850 nm vs 940 nm IR Lights: What is the Difference?](#) (850 nm offers greater range and sensitivity for cameras) ↵
7. [Axton Tech – 850 nm vs 940 nm IR Lights: What is the Difference?](#) (940 nm has ~50% shorter effective range vs 850 nm) ↵
8. [Stanley Electric – Case Study: Infrared LEDs in TOA “TRIFORA” Security Cameras](#) (high-power 940 nm LED arrays for 0 lux surveillance) ↵
9. [Christie Medical – VeinViewer Illumination Technology](#) (near-IR light revealing subcutaneous veins) ↵
10. [OmniVision – Nyxel® NIR Image Sensor Technology](#) (enhanced 940 nm sensitivity for driver monitoring, etc.) ↵
11. [Lasernet – Overview of LED/Laser Classification \(IEC 60825-1\)](#) (LEDs are included in laser safety standards; class 1 criteria) ↵
12. [Vishay – Eye Safety of Infrared Optical Emitters \(App Note\)](#) (all Vishay IR LEDs fall under Class 1 / Exempt, arrays need caution) ↵
13. [Tech-LED – SMT850-251 850 nm LED Datasheet](#) (fast switching: ~100 ns rise/fall time) ↵
14. [Tech-LED – SMT850D Product Page](#) (enhanced 850 nm LED, supports 100 mA continuous drive) ↵
15. [Tech-LED – L365-66-60 60-Chip UV LED Array Datasheet](#) (reference: ~130 mW radiant flux for 60-die UV LED) ↵

16. [Axton Tech – 850 nm vs 940 nm IR Lights: What is the Difference?](#) (940 nm LED has ~40% of the radiant intensity of 850 nm) ↵
17. [Axton Tech – 850 nm vs 940 nm IR Lights: What is the Difference?](#) (typical silicon camera sensors are much less sensitive at 940 nm vs 850 nm) ↵
18. [Freestyle Systems – LEDs and Infrared Light](#) (visible LEDs emit negligible IR; benefit for salons/museums) ↵
19. [Moon LEDs – The Ultimate Guide to IR LED](#) (IR LEDs offer high electrical-to-IR conversion efficiency) ↵
20. [Moon LEDs – The Ultimate Guide to IR LED](#) (IR LEDs in remote controls for TVs, etc.) ↵
21. [Moon LEDs – The Ultimate Guide to IR LED](#) (IR LEDs in security cameras – invisible illumination detectable by camera sensors) ↵
22. [Moon LEDs – The Ultimate Guide to IR LED](#) (IR LED emits photons via recombination just like a normal LED) ↵



The Tech-LED team provides deep technical guidance on LEDs and optoelectronics, including design tips, industrial application notes, and updates on emerging wavelengths and packages.